



Robotiq Adaptive Grippers on Polyscope X - User Guide

Original instructions (en)

© 2026 Robotiq Inc.

START PRODUCTION FASTER

Table of Contents

Revisions.....	3
Known Issues (beta version UCX-3.28.0).....	5
1. Product Compatibility on Polyscope X.....	6
Current Feature Set for Adaptive Grippers.....	6
2. Main differences between Polyscope 5 & Polyscope X.....	7
a. Tool I/O: Manual scan to detect gripper.....	7
b. 1 application = 1 program.....	9
c. Start program - No confirmation step.....	9
d. Other differences are minor and relate to user interface interactions.....	9
3. Configuring Polyscope X for Robotiq Adaptive Grippers on Tool Port.....	10
a. Configuration (Initial Setup for New Program).....	10
b. Robot Power-Up.....	14
c. Sidebar (Right Side).....	16
4. Adding a Node.....	17
a. Gripper Activate Node.....	18
b. Gripper Grip Check Node.....	19
c. Gripper Move Node.....	21
5. Gripper Activation From ToolBar.....	22
6. Open and Close Grippers Smart Skills.....	25
a. Insert Open and Close Nodes.....	25
b. Modifying the Default Parameters for the Gripper Open and Gripper Close Smart Skills.....	29

Revisions

Robotiq may modify this product without notice, when necessary, due to product improvements, modifications or changes in specifications. If such modification is made, the manual will also be revised, see revision information.

2026/01/30

[BREAKING CHANGE] Update to support Polyscope X version 10.12.0

[NEW] Add Gripper Close and Gripper Open Smart Skills.

[NEW] Gripper power output and communication parameters are now set automatically when selecting "Robotiq Adaptive Gripper" in Application->Communication->Tool I/O menu.

[NEW] Gripper scans can now be triggered from the sidebar.

[NEW] Grippers can now be activated directly from the sidebar.

2025/05/13

[BREAKING CHANGE] Update to support Polyscope X version 10.8.0

[NEW] Added the capability to jog multiple grippers from the new Robotiq Sidebar.

2025/04/23

[NEW] Add the If and Else nodes as child nodes to the Gripper Grip Check node to properly handle object detection.

[NEW] French translation is now available

[NEW] Multiple grippers can be operated simultaneously in the gripper move node.

[NEW] Added the capability to see the gripper information in the Adaptive grippers page.

[NEW] Added the capability to update the firmware of the Grippers

[BUGFIX] Fixed a bug where object detection status could be wrong in the Edit Gripper Action page.

2025/03/17

[BUGFIX] Fixed an issue when the Gripper ID 1 is used with the Grip Check node to validate a dimension.

2025/03/05

[IMPROVEMENT] Renamed URCapX file to match other Robotiq URCaps' naming pattern.

[BUGFIX] Fixed graphical glitch where the keyboard would hide the text fields when configuring dimension validation in Grip Check.

[BUGFIX] Fixed bug where dimension validation in Grip Check could be configured even when "Not detected" was selected.

[BUGFIX] Fixed bug where Gripper Move's "Go to position" button would not be enabled at the right moments.

[BUGFIX] Removed non-functional Smart Skill.

2025/03/04

Initial release

Known Issues (beta version UCX-3.28.0)

- [KNOWN ISSUE] Re-installing the URCapX may cause the gripper to lose communication. A manual restart (Shutdown -> Restart) of the robot after the URCapX change fixes that.

1. Product Compatibility on Polyscope X

Product	Current status
Hand-E, Hand-E C10, 2F-85, 2F-140	✓ (beta)
EPick, PowerPick	Not supported
Other Plug & Play components	Not supported
Palletizing Solution	Not supported

Table 1-1: Robotiq Products: Supported and Unsupported

Current Feature Set for Adaptive Grippers

Feature	Polyscope X (Feb 2026)	Polyscope 5
Gripper activation	✓	✓
Gripper open/close/move	✓	✓
Grip check	✓	✓
Script programming	✓	✓
Multi-Grippers		✓
Toolbar	✓	✓
Stroke calibration		✓
Robot compliance		✓

Table 1-2: Features of Adaptive Grippers for Polyscope 5 and Polyscope X

2. Main differences between Polyscope 5 & Polyscope X

a. Tool I/O: Manual scan to detect gripper

With Polyscope X, when the gripper is connected to the robot's wrist tool port, you must manually scan for the gripper presence through the user interface.

Note: If the gripper is connected to a USB port, this step can be skipped.

During the scan, 24 volts will be sent to the gripper to configure communication parameters. When starting the robot with the gripper connected to the wrist, you must manually initiate a scan to establish communication between the software and the gripper. Auto-scanning is avoided to prevent hardware damage if a different tool is connected instead of the Robotiq gripper. The robot must be activated for this process, and the gripper will display once communication is established.

1. In the "Application" tab, choose the "Robotiq" tile.

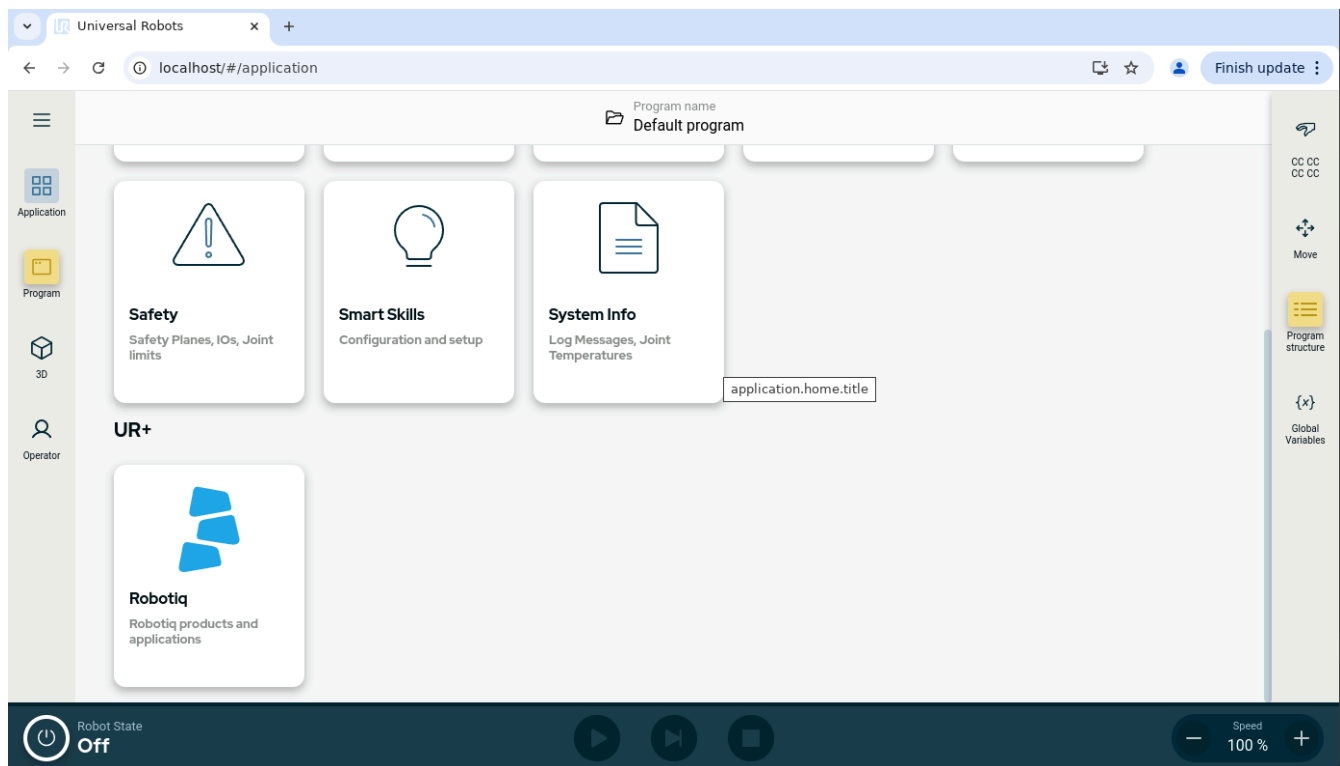


Fig. 2-1: Application Tab - Robotiq

Step 2. In the "Adaptive Grippers Dashboard," select the "Gripper is connected to wrist" checkbox.

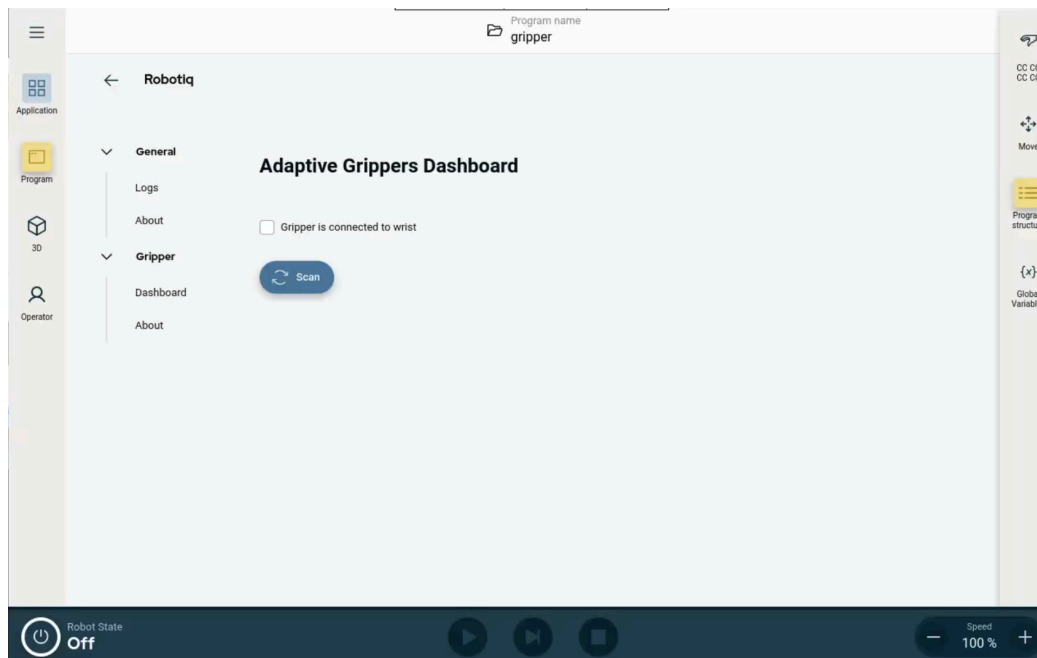


Fig. 2-2: Adaptive Grippers Dashboard

Step 3. Activate the Robot.

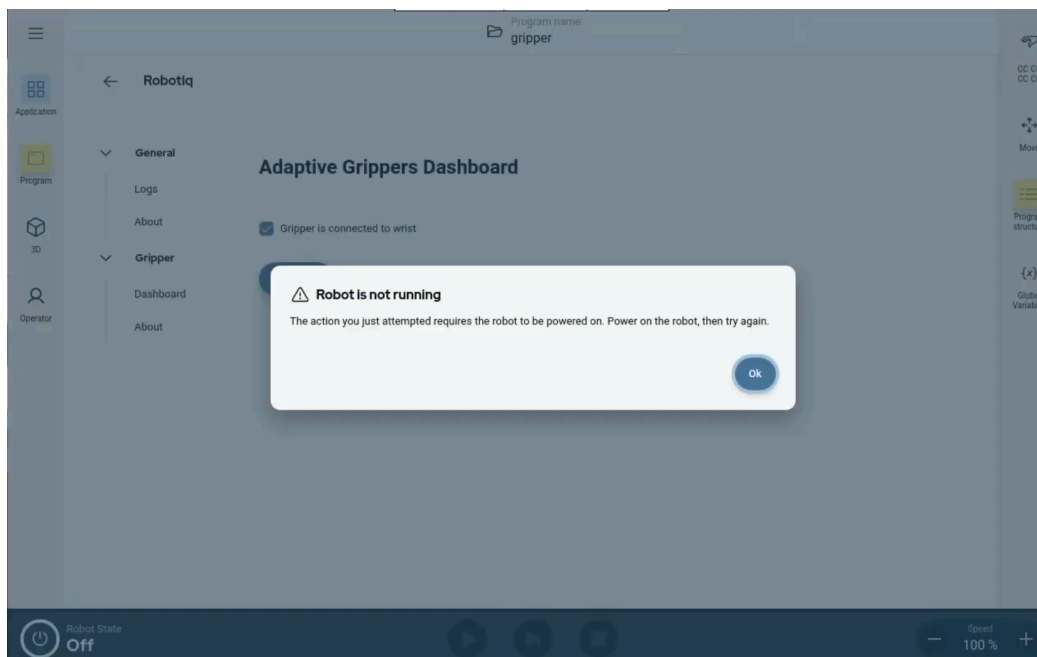


Fig. 2-3: Robot is not running Window

Step 4. Click the "Scan" button.

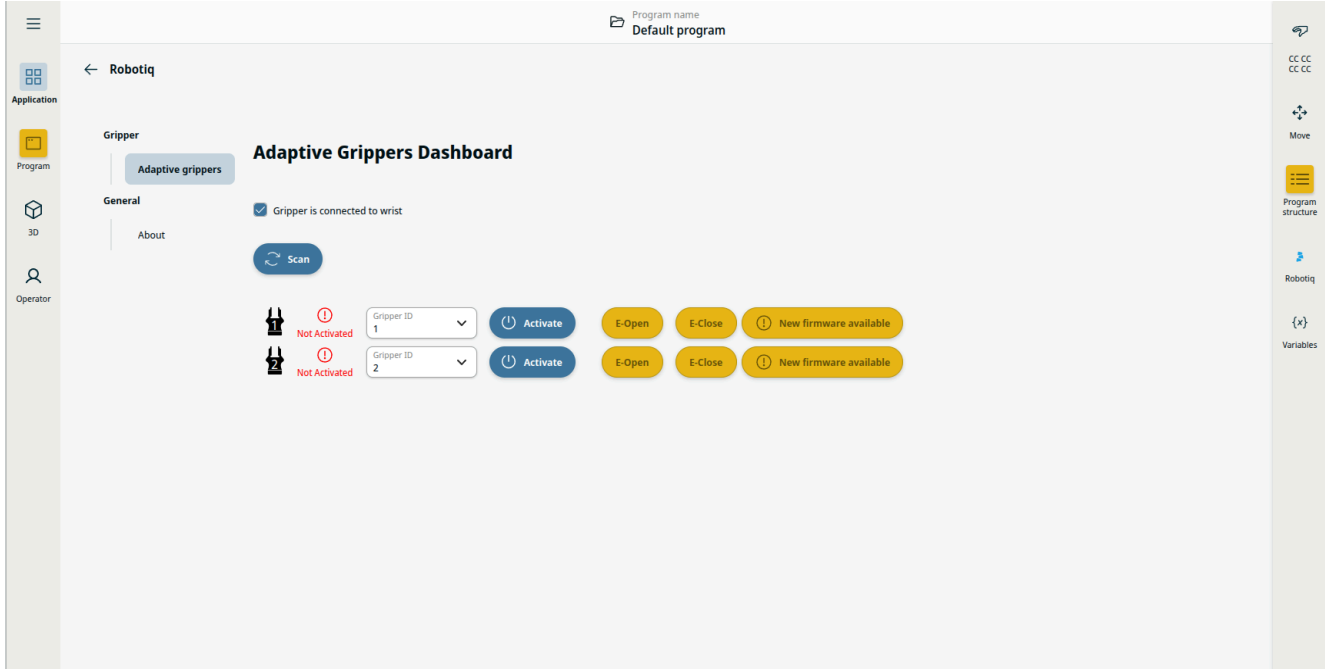


Fig. 2-3: Adaptive Grippers Dashboard - Scan

Step 5. Three Options to Activate the Gripper:

- Option 1. Use the activate button next to the gripper entry in the gripper list.
- Option 2. Click the "Activate" button in the "Gripper Activate" program node.
- Option 3. Start a program with a "Gripper Activate" program node.

b. 1 application = 1 program

One program corresponds to one application. When a new program is created, it generates a new installation node with its own parameters.

c. Start program - No confirmation step

Important: Polyscope X starts programs immediately without any confirmation prompts. It doesn't ask where to start the program or if you want to move the robot to the start position. Unlike Polyscope 5, Polyscope X skips this additional confirmation step for security, so be sure to keep this difference in mind.

d. Other differences are minor and relate to user interface interactions.

3. Configuring Polyscope X for Robotiq Adaptive Grippers on Tool Port

a. Configuration (Initial Setup for New Program)

Step 1. On the left panel, select “Application”.

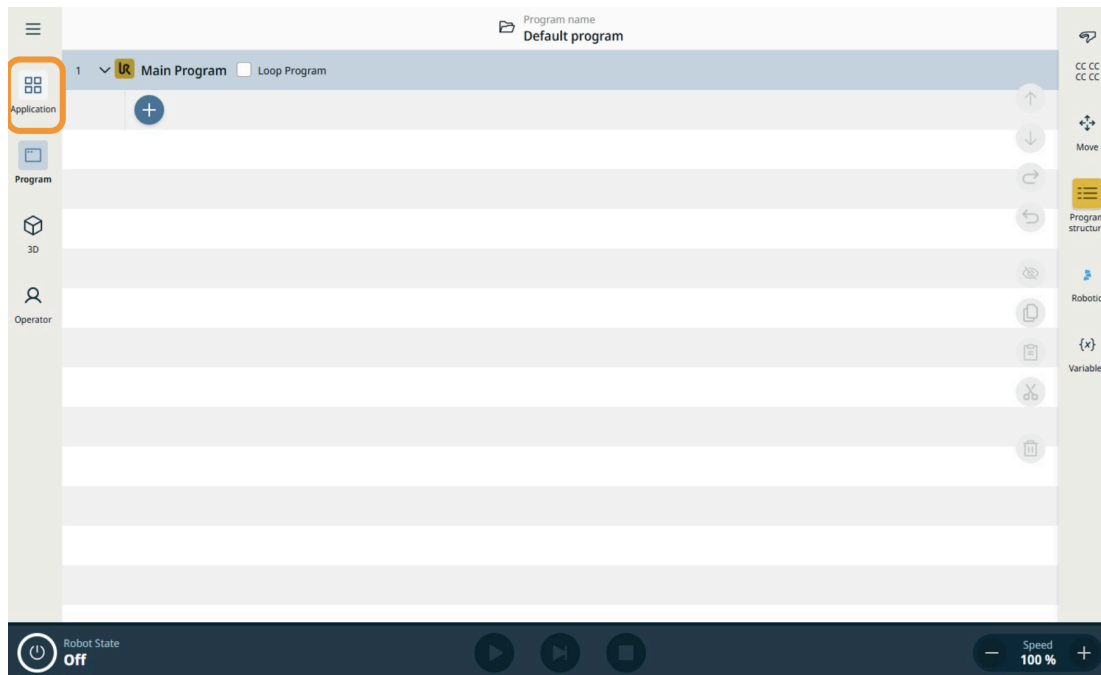


Fig. 3-1: New Program

Step 2. Select "Robotiq" within the "URCaps" section.

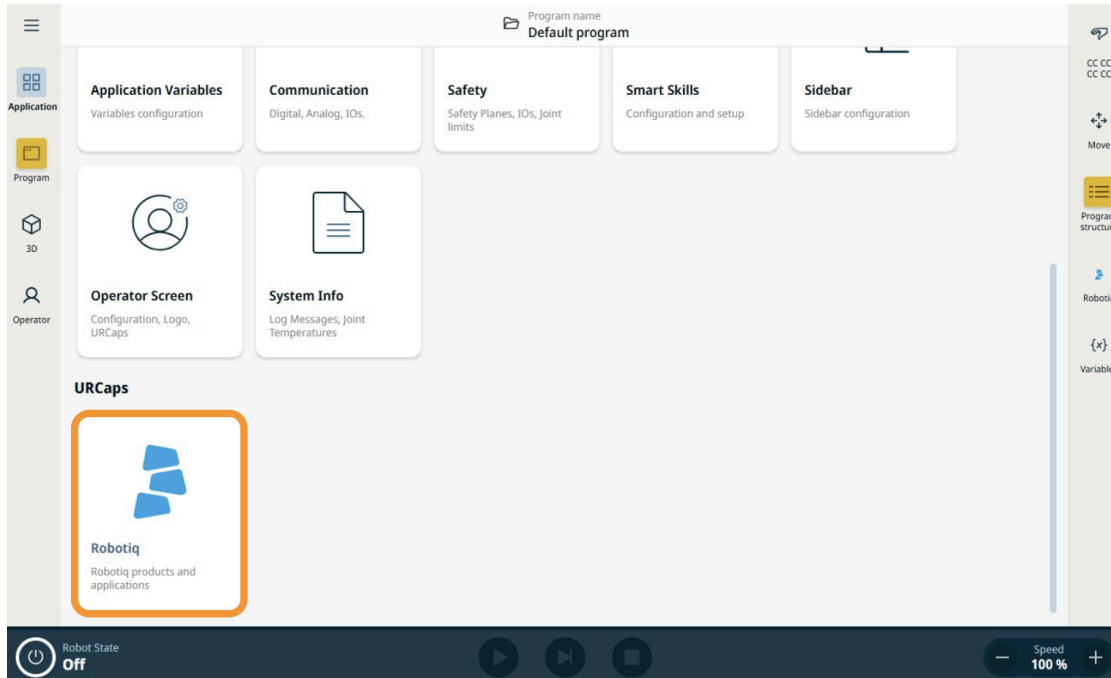


Fig. 3-2: URCaps Section

Step 3. Ensure that the "Gripper is connected to wrist" checkbox is selected within the "Adaptive Grippers" tab.

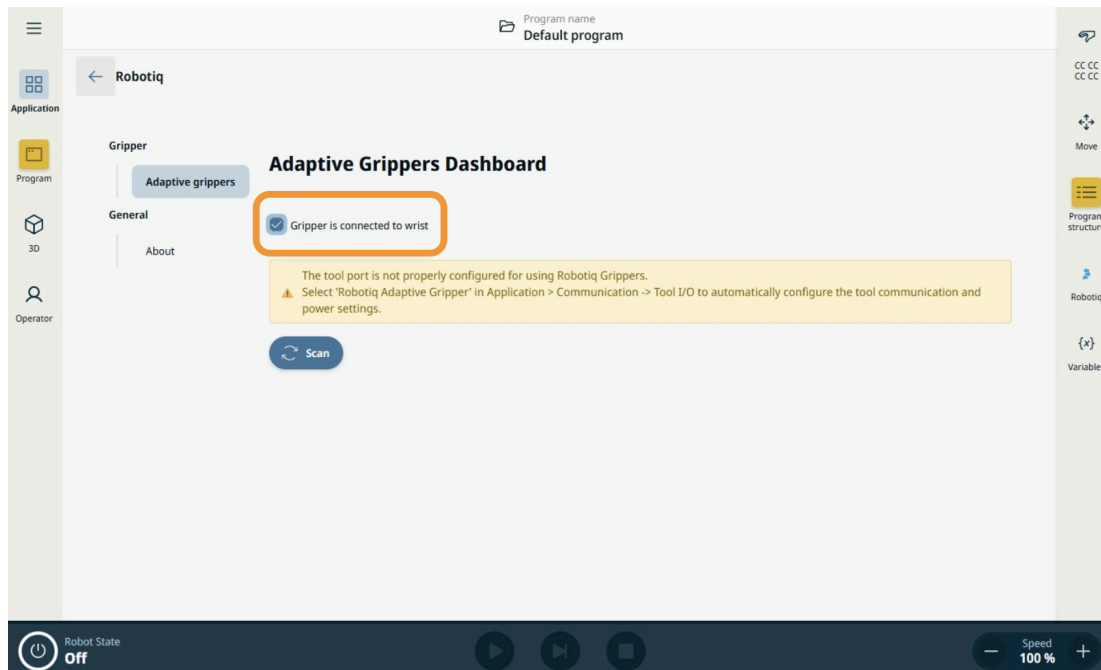


Fig. 3-3: Adaptive Grippers Dashboard

Step 4. Return to the "Application" menu.

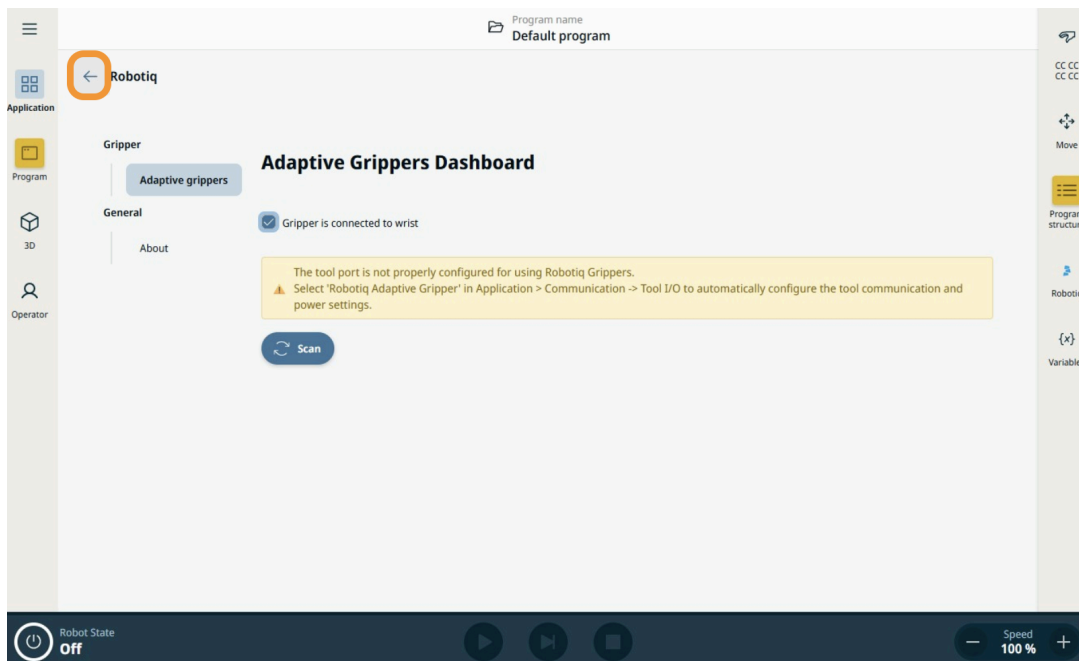


Fig. 3-4: Adaptive Grippers Dashboard

Step 5. Under the "Application" menu, choose "Communication".

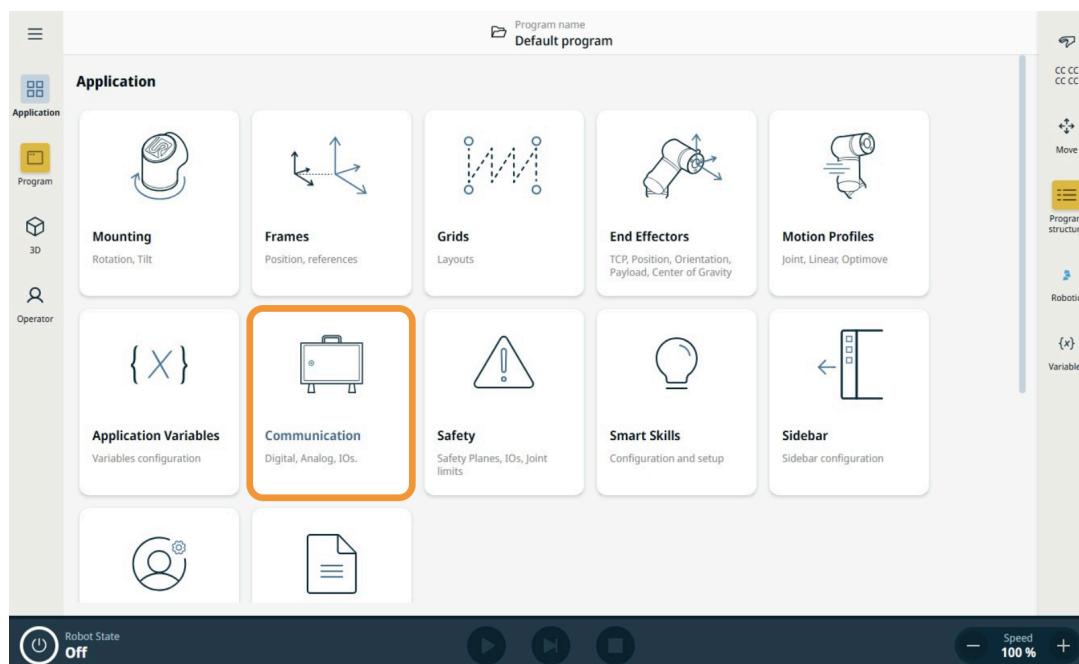


Fig. 3-5: Application Menu

Step 6. Under the "Robot" tab, choose "Tool I/O".

Step 7. Open the "Controlled By" drop-down.

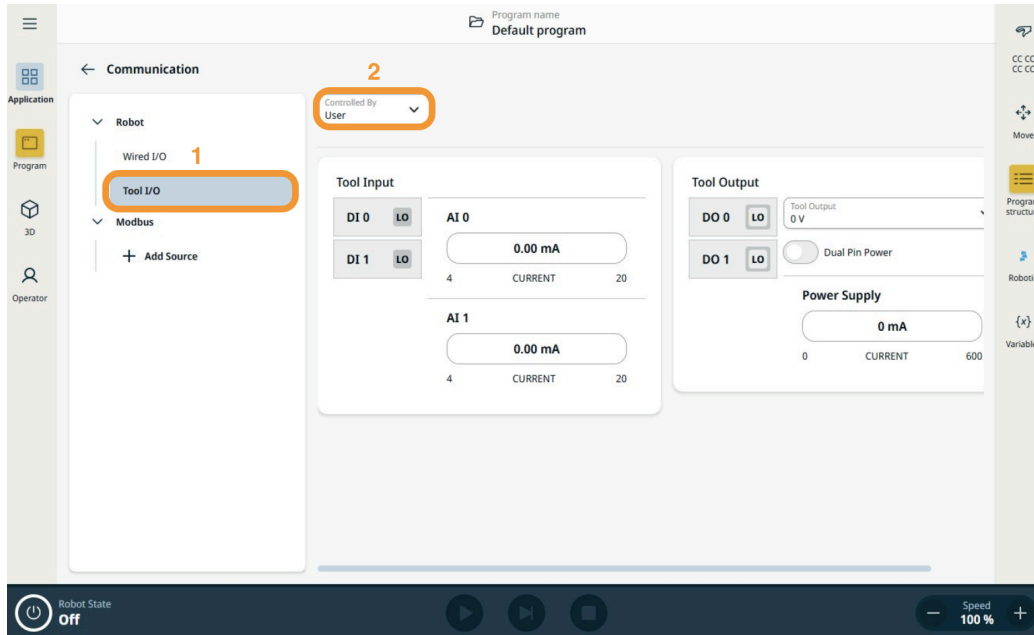


Fig. 3-6: Controlled By Drop-down (Closed)

Step 8. Select the "Robotiq Adaptive Gripper" from the drop-down menu.

Note: This item appears only when the "Gripper is connected to Wrist" option is selected in the Robotiq application menu.

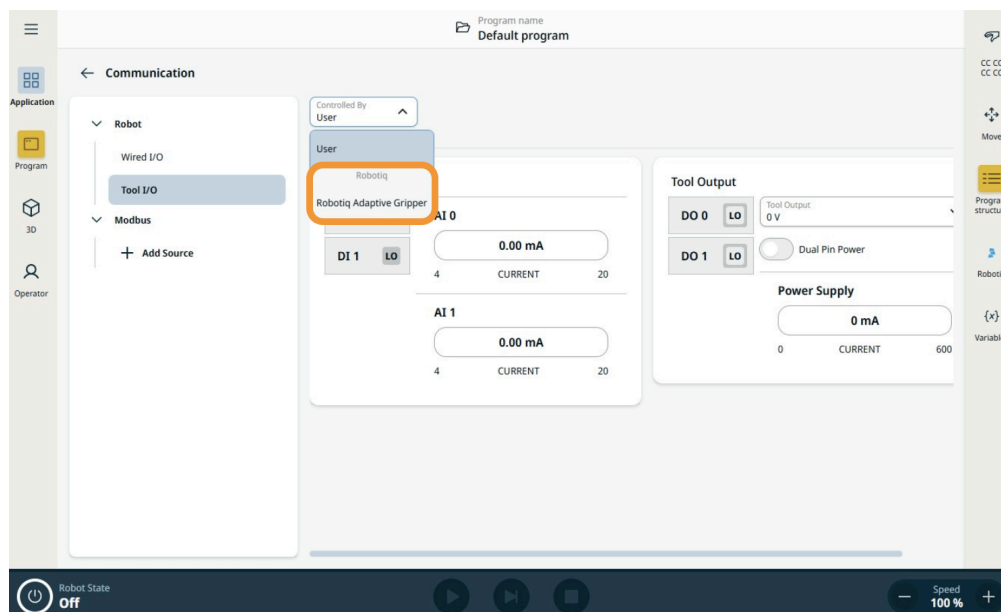


Fig. 3-7: Controlled By Drop-down (Expanded)

Step 9. Return to the "Application" menu.

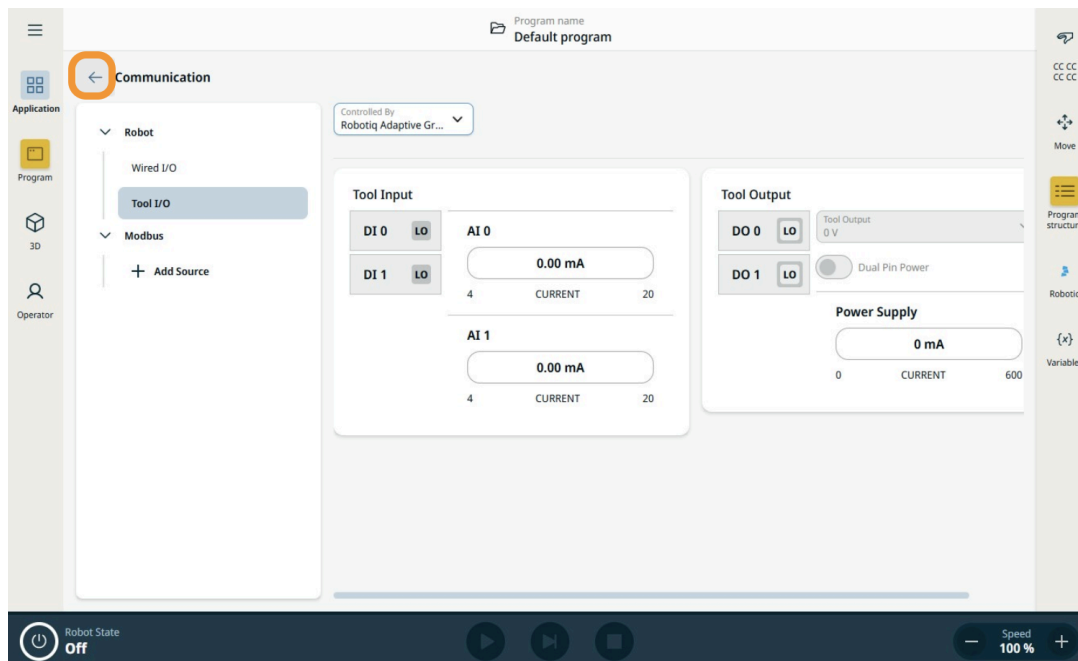


Fig. 3-8: Communication Menu

b. Robot Power-Up

Step 1. Click the "Power" icon located in the bottom-left corner of the screen.

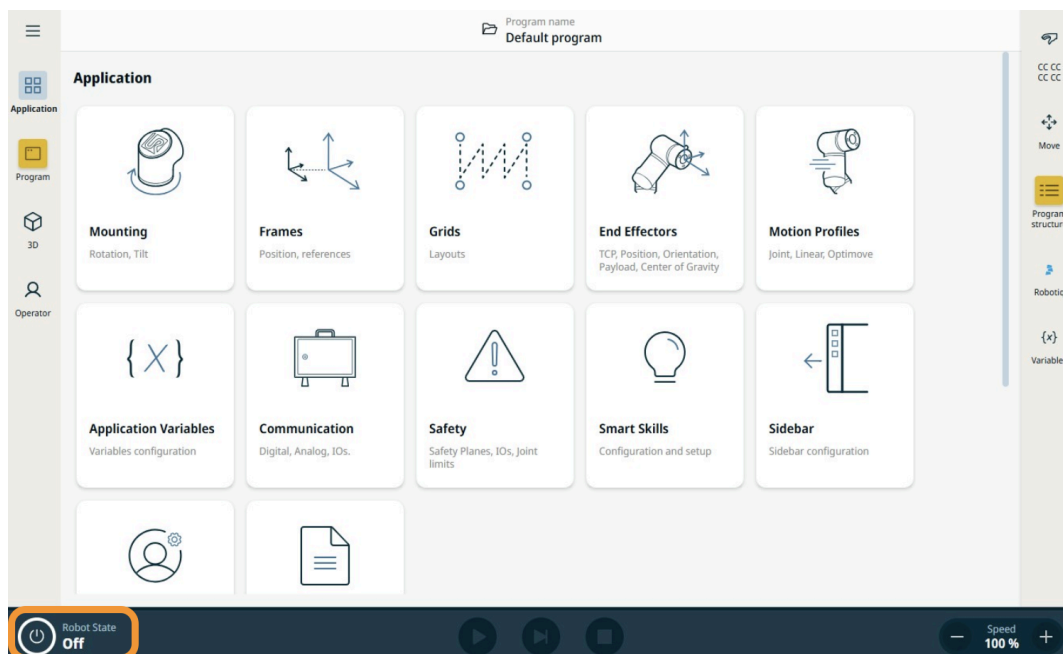


Fig. 3-9: Application Menu - Power Icon

Step 2. Select the "Power On" button.

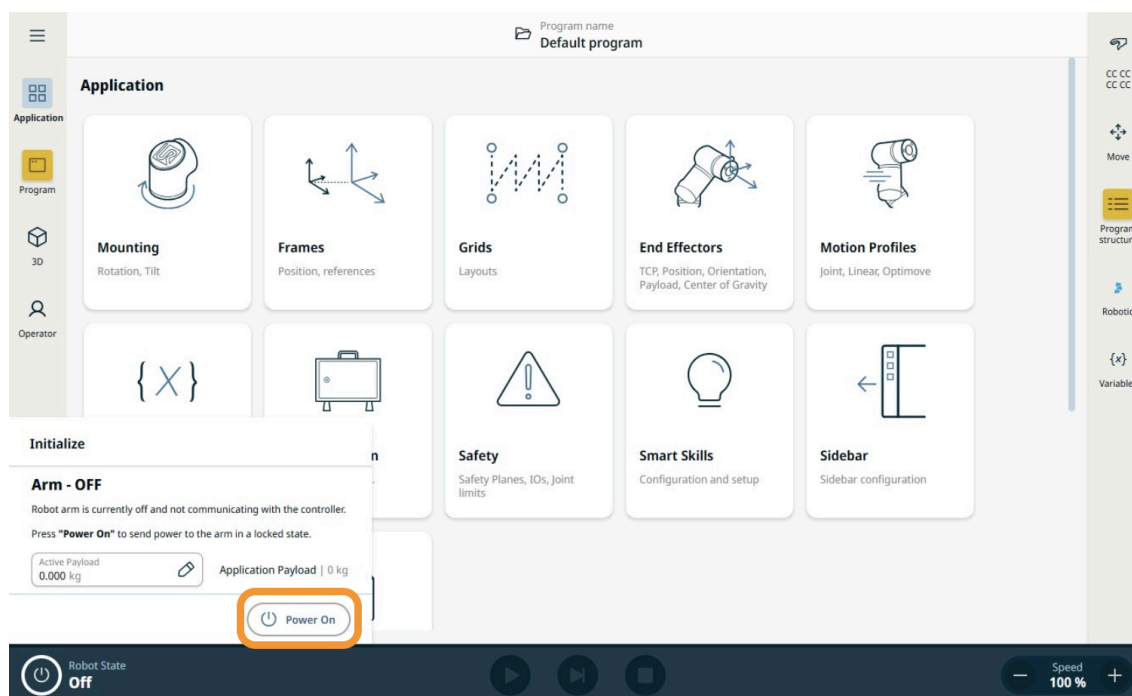


Fig. 3-10: Initialize Window - Arm OFF

Step 3. Select "Unlock".

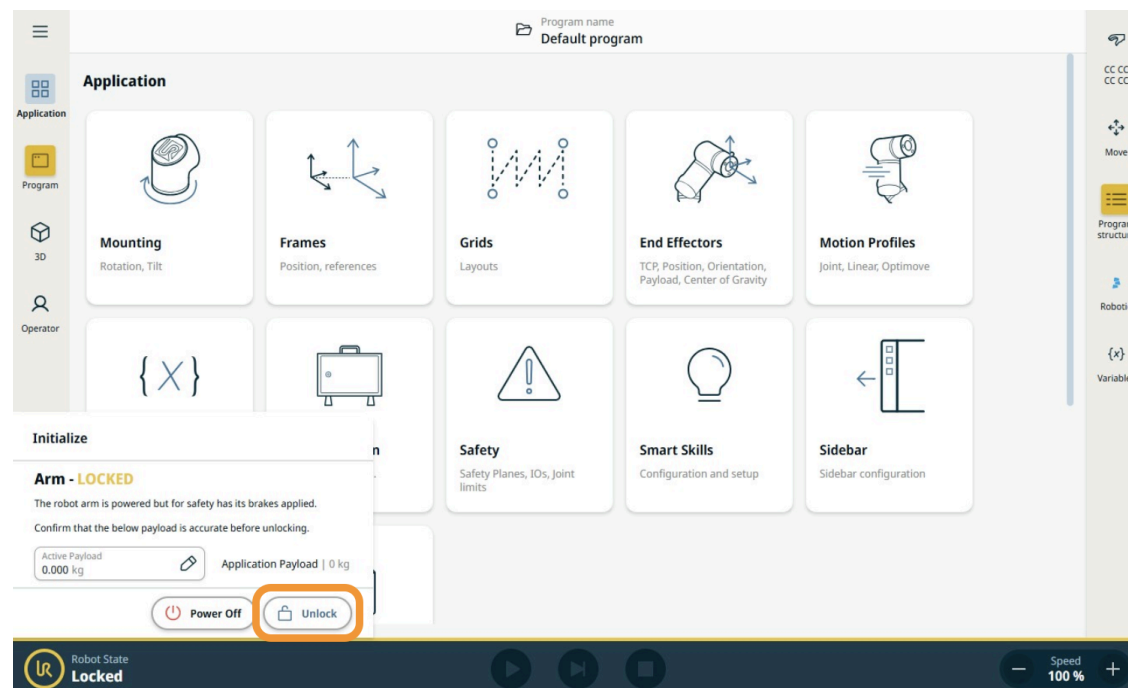


Fig. 3-11: Initialize Window - Arm LOCKED

c. Sidebar (Right Side)

Step 1. Select "Robotiq" from the side bar on the right side of the screen.

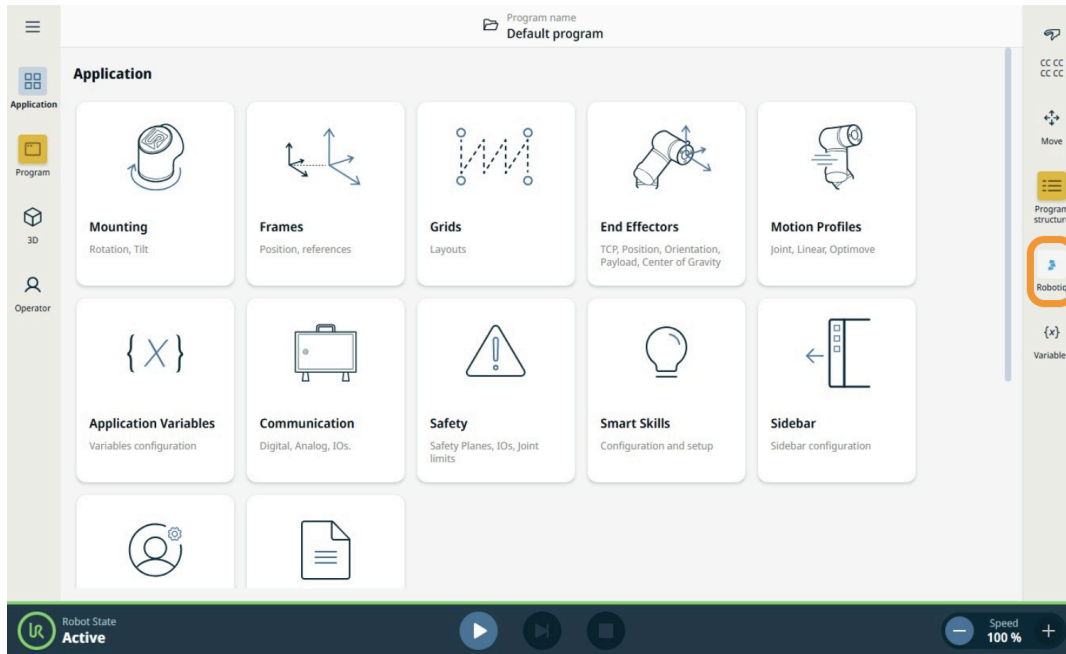


Fig. 3-12: Application Menu - Side Bar (Right Side)

Step 2. Detected grippers will instantly appear in the "Robotiq" menu of the sidebar. If any grippers are not listed, select the "Scan" button to attempt detection again.

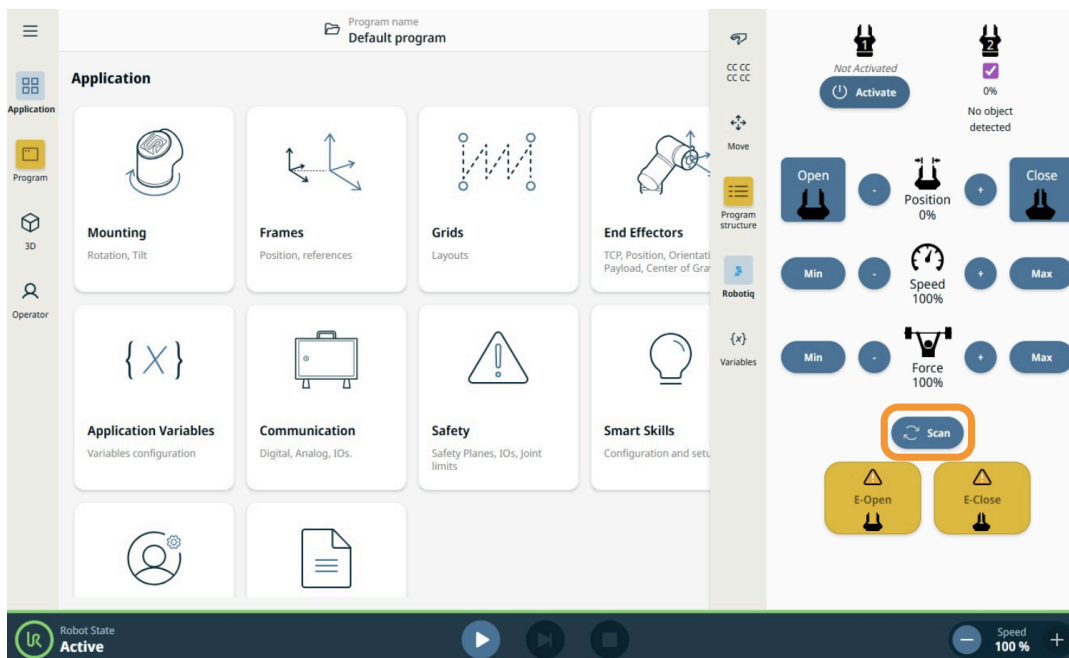


Fig. 3-13: Robotiq Menu (Side Bar)

4. Adding a Node

Step 1. From the program, select the  button to add a node. A window will appear, showing a list of available node types.

The URCap nodes include:

- Gripper Activate
- Gripper Grip Check
- Gripper Move

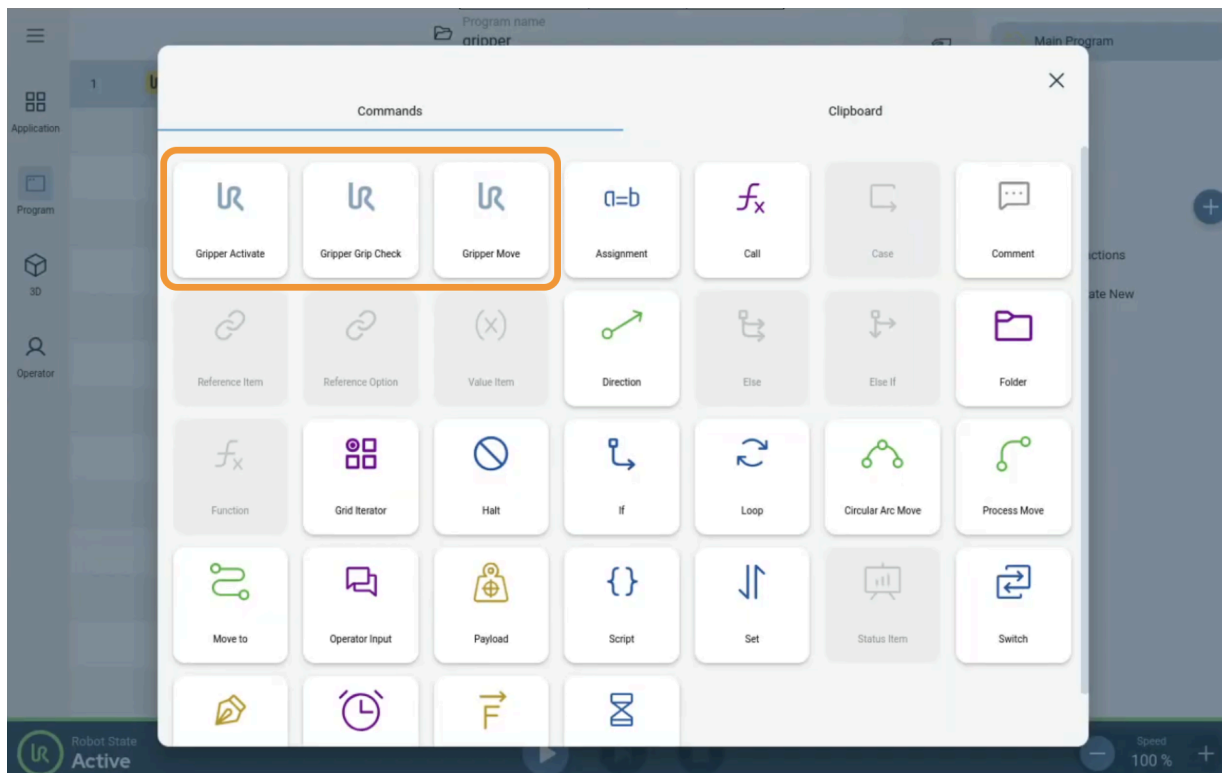


Fig. 4-1: Nodes List

a. Gripper Activate Node

When you select the “Gripper Activate Node”, the “Configure Gripper Activate” window will appear with two options:

- **Activate**

Functions the same as in Polyscope 5. If the gripper is already activated, the system will not reactivate it, and the program will proceed to the next line. If the gripper is not activated, the system will activate it.

- **Reset and Activation**

Restarts a complete gripper activation sequence, removing any errors. This node works exactly the same as the Polyscope 5 “Gripper Activate” program node.

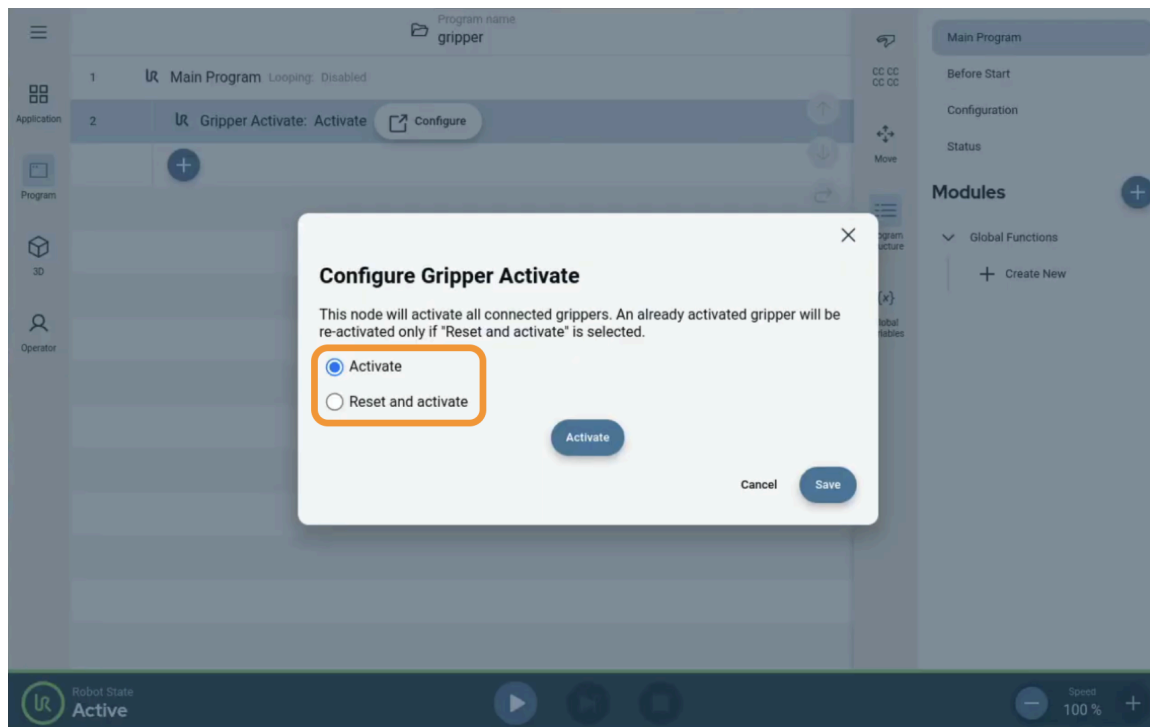


Fig. 4-2: Configure Gripper Activate Window

b. Gripper Grip Check Node

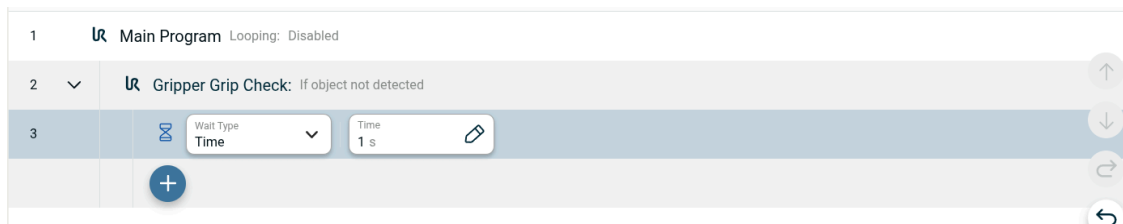


Fig. 4-3: Gripper Grip Check Node - If object not detected

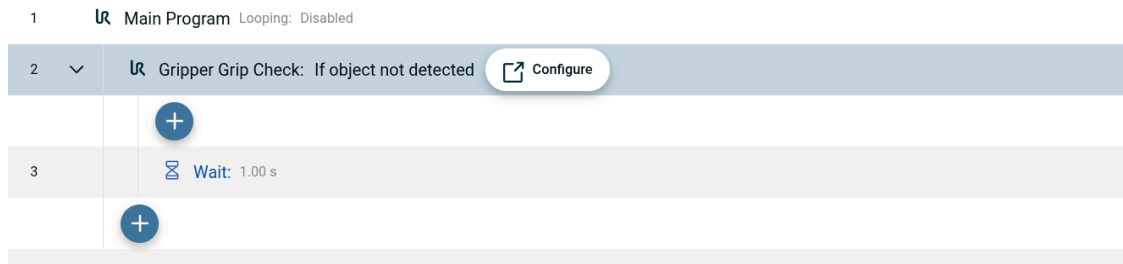


Fig. 4-4: Gripper Grip Check Node - If object not detected

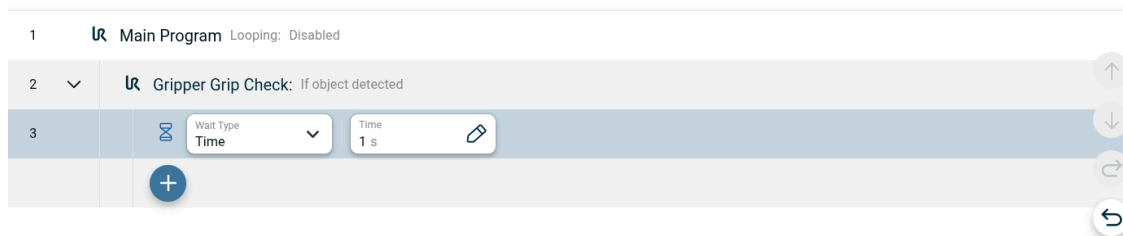


Fig. 4-5: Gripper Grip Check Node - If object detected

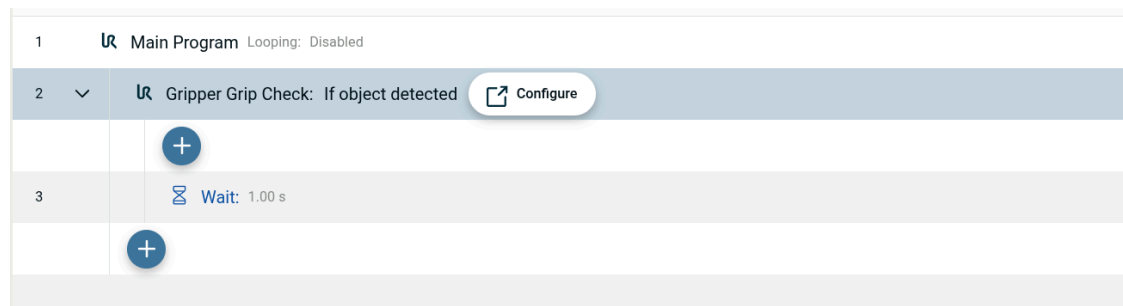


Fig. 4-6: Gripper Grip Check Node - If object detected

Step 1. Selecting "Configure" opens the "Configure Grip Check" window.

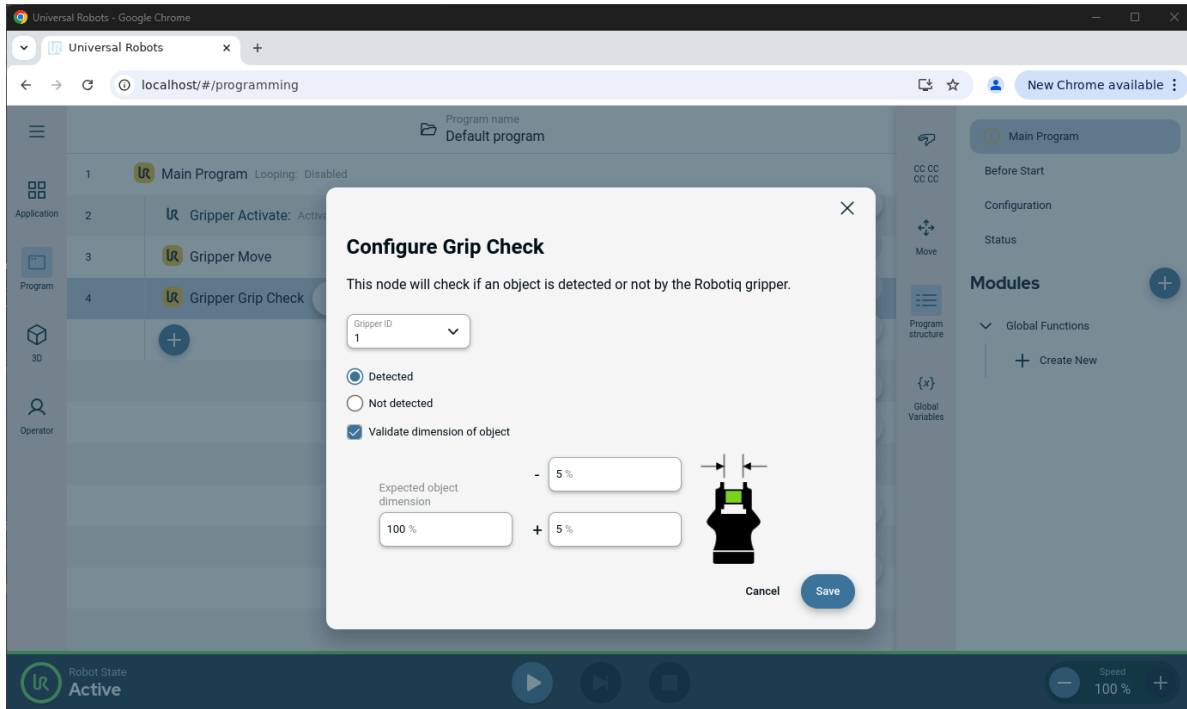


Fig. 4-7: Configure Grip Check Window

Step 2. Adjust the parameters

- Gripper ID: If more than one gripper is present, you can select which gripper ID to use for the grip check.
- Validate dimension of object:
 - Gripper dimension is measured as a percentage of closure. 0% indicates fully open, while 100% indicates fully closed.
 - Calibration of dimension (specifying units in mm or inches as in Polyscope 5) is not yet available.

Notes: Child nodes of the grip check node are only executed if the condition is met. The node works as it does in Polyscope 5.

c. Gripper Move Node



Fig. 4-8: Gripper Move Test & Configuration

Step 1. Configure Gripper Move

The "Configure Gripper Move" window will open when the node is selected.

Step 2. Edit Gripper Action

To modify the gripper's settings, choose "Edit action." This will open the "Edit Gripper Action" window, allowing you to set the desired position, speed, and force.

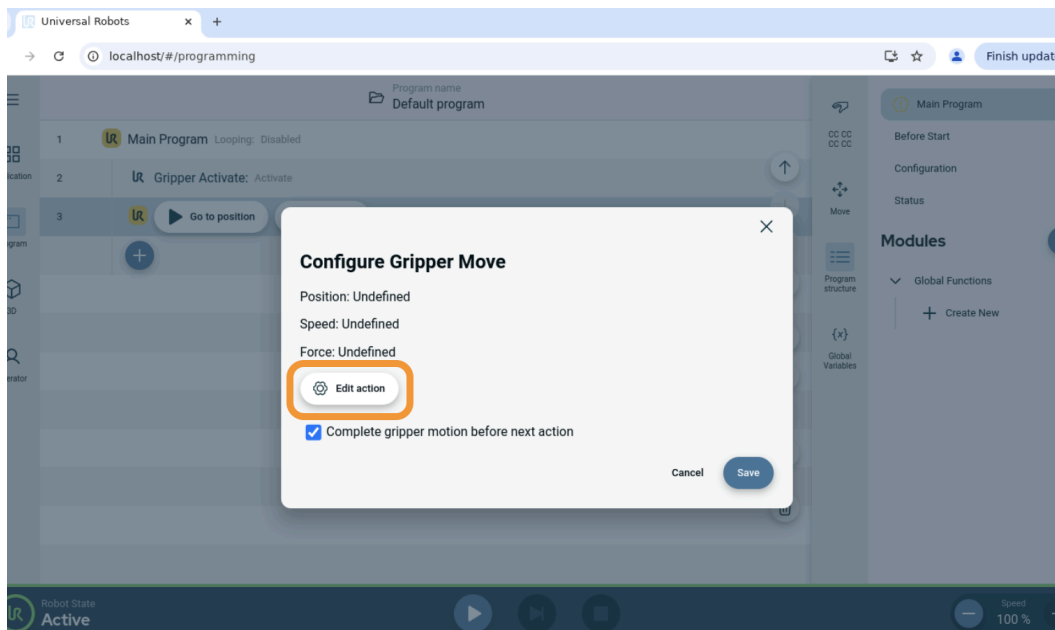


Fig. 4-9: Configure Gripper Move Window

Notes:

- Currently, only a single gripper is supported. If multiple grippers are present, the first activated gripper in the list will be used.
- The "Go to Position" button moves the gripper to the taught position.
- The "Configure" button opens a popup with details about the current node configuration.
- The "Edit Action" button is used to teach the gripper's position. The "Edit Action" window works the same way as in Polyscope 5.
- Compliance mode is not available in the PSX URCapX.

5. Gripper Activation From ToolBar

Step 1: Under the "Application" menu, choose "Robotiq" within the URCaps section.

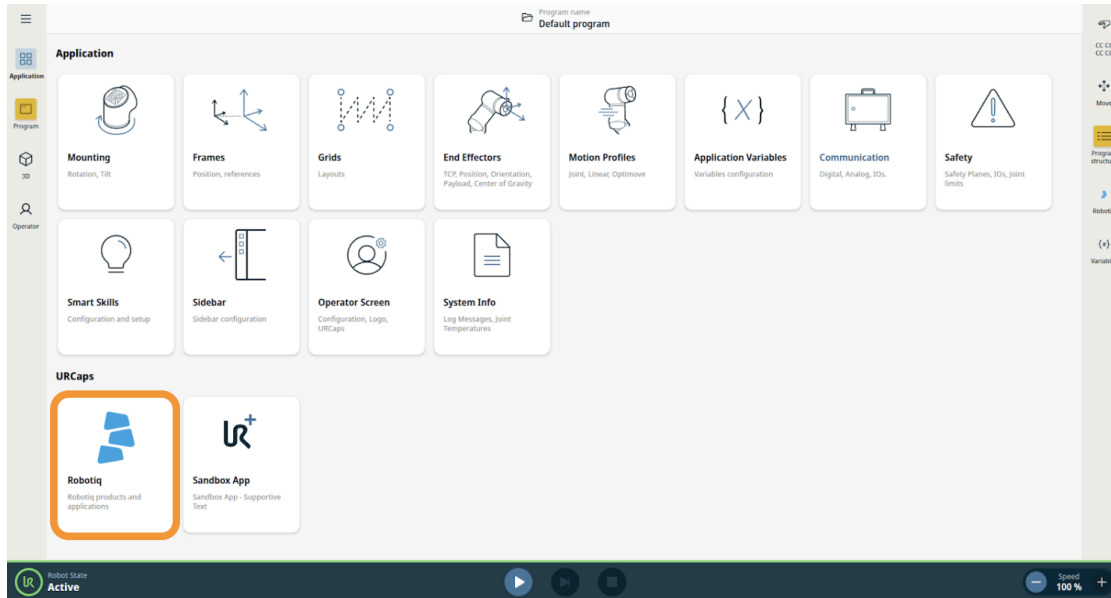


Fig. 5-1: Application Menu - URCaps Section

Step 2: The available grippers are listed for activation within the "Adaptive Grippers Dashboard" located under the "Adaptive Grippers" tab.

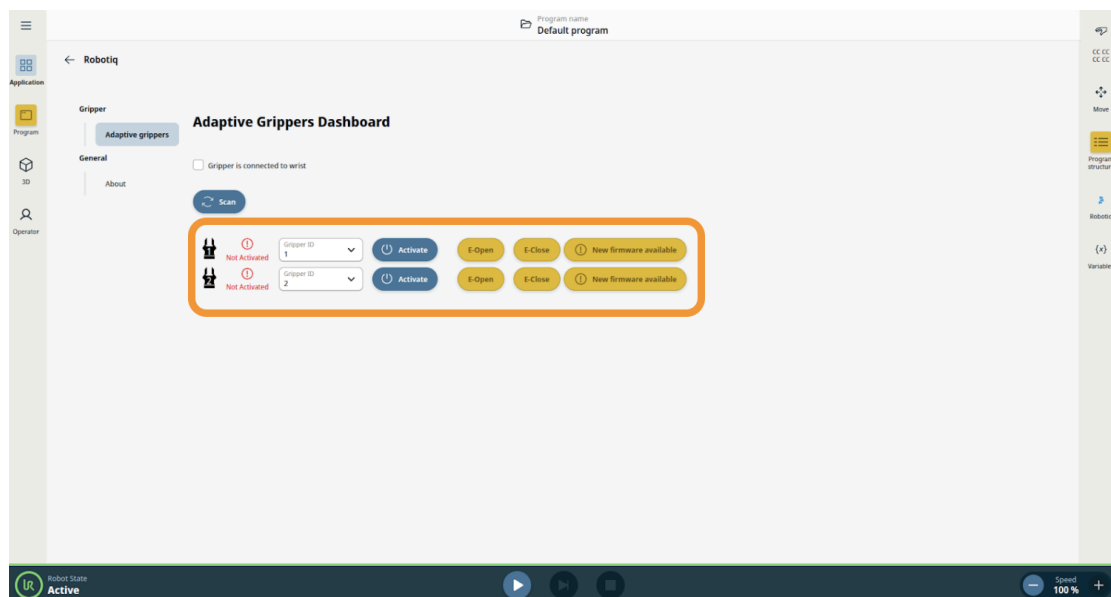


Fig. 5-2: Adaptive Grippers Dashboard - List of Available Grippers

Step 4: To activate a gripper, open the toolbar by selecting the "Robotiq" tab located in the right sidebar.

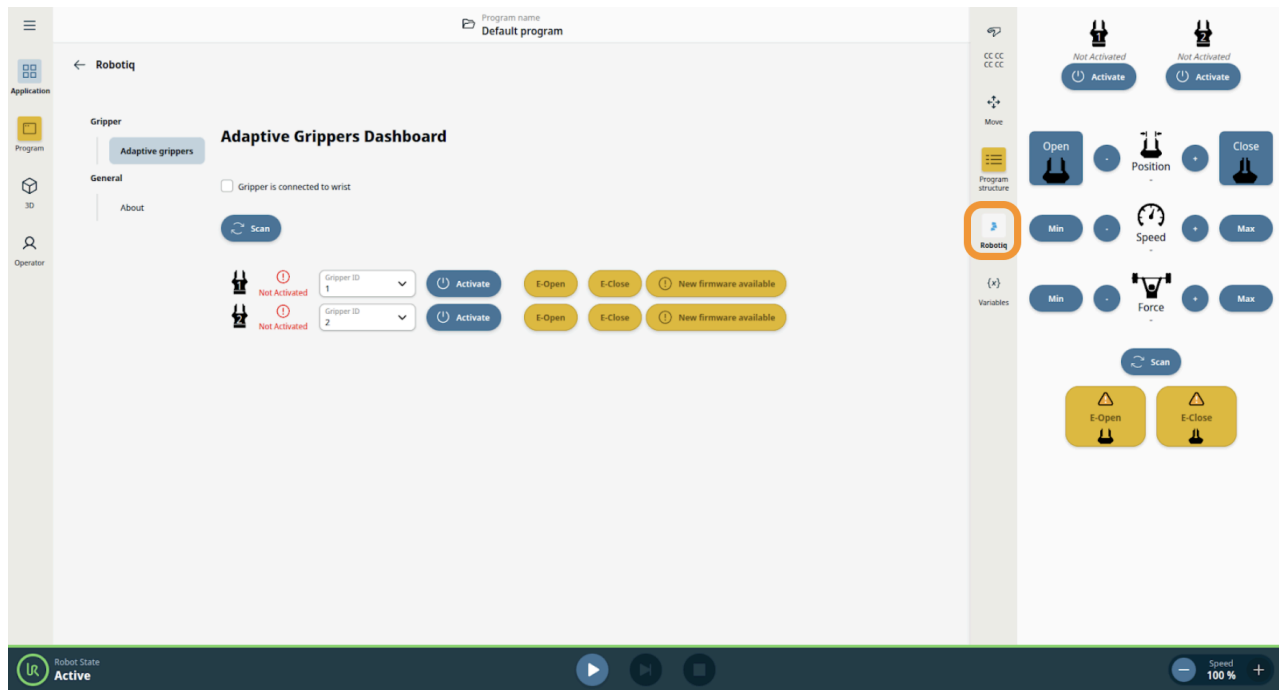


Fig. 5-3: Robotiq Menu (Right Sidebar)

Step 5: Identify any grippers (e.g., Gripper 1, Gripper 2) that have not been activated.

Step 6: To activate the desired gripper, press the "Activate" button.

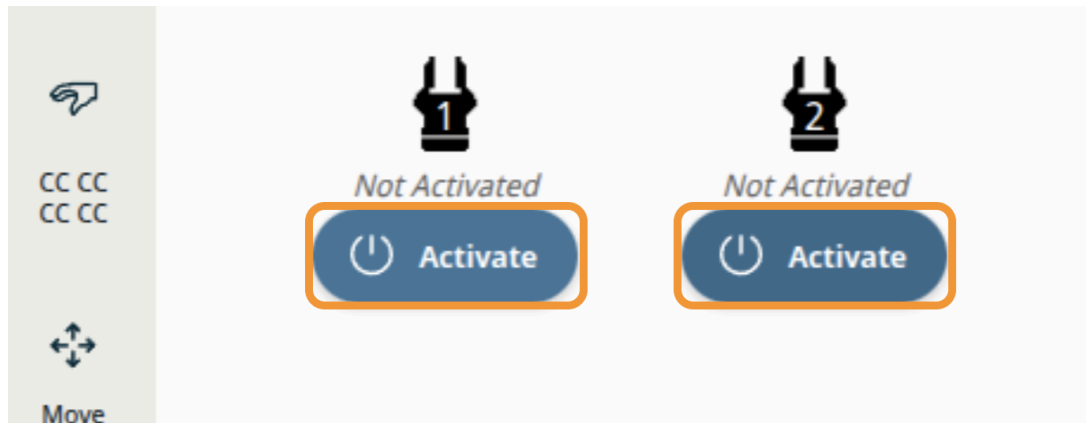


Fig. 5-4: Robotiq Menu (Right Sidebar) - Zoomed View

Step 7: Check the activation status. Look for the status update in the toolbar and confirm that the activation is successful.

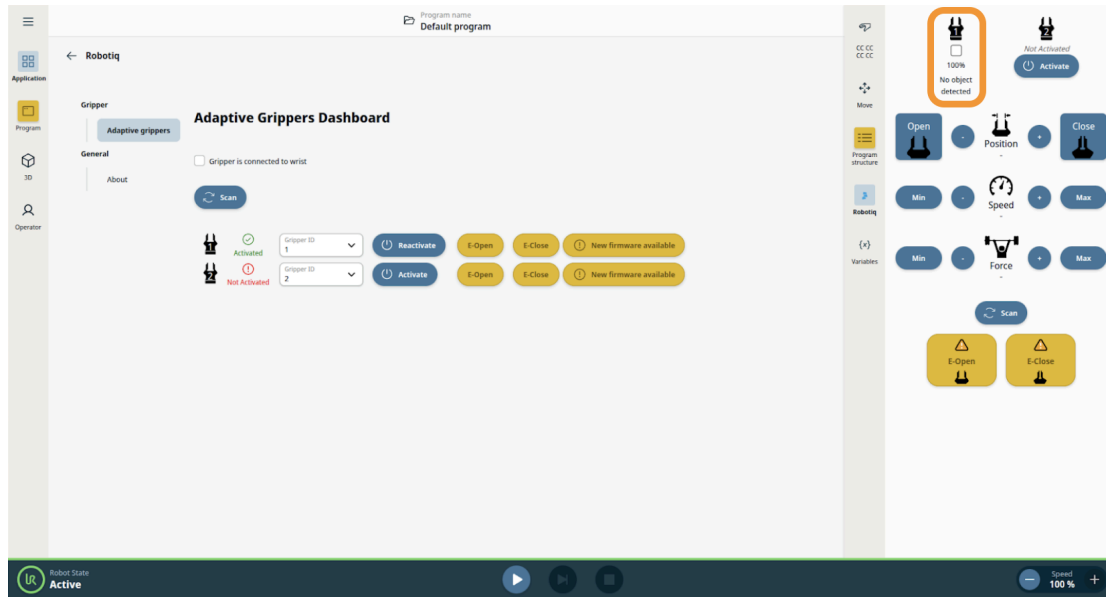


Fig. 5-5: Robotiq Menu (Right Sidebar) - Activation Status

Step 8: Use the open and close buttons to control the gripper's position. Confirm that the position is shown as a percentage.

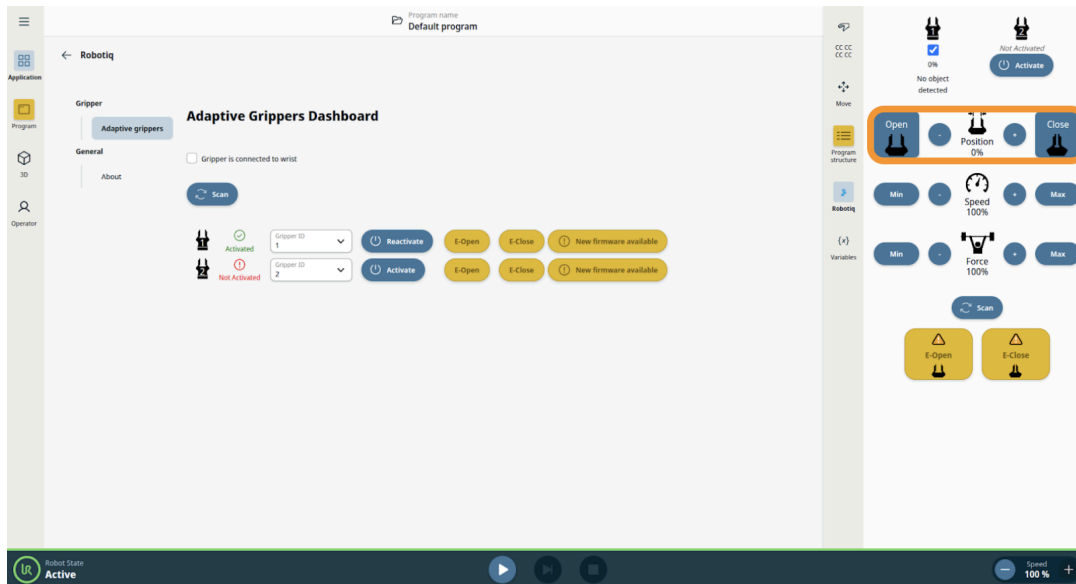


Fig. 5-6: Robotiq Menu (Right Sidebar) - Open & Close Buttons

6. Open and Close Grippers Smart Skills

a. Insert Open and Close Nodes

Step 1: On the left sidebar, choose "Program".

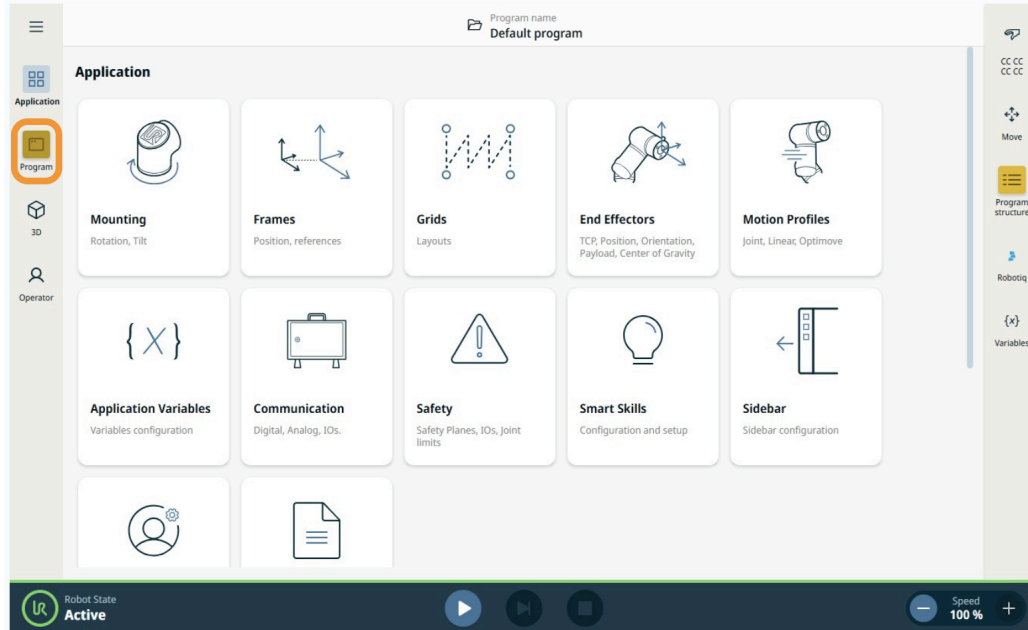


Fig. 6-1: Program - Left Sidebar

Step 2: On the right sidebar, click the "Move" tab.

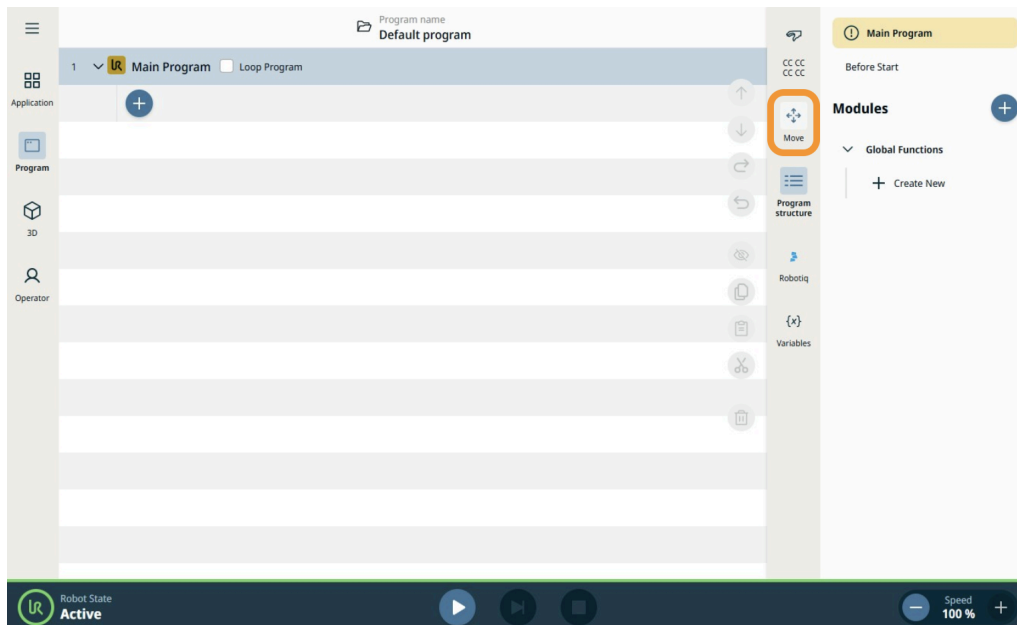


Fig. 6-2: Move Menu - Right Sidebar

Step 3: Select "Smart Skills" at the top of the "Move" section.

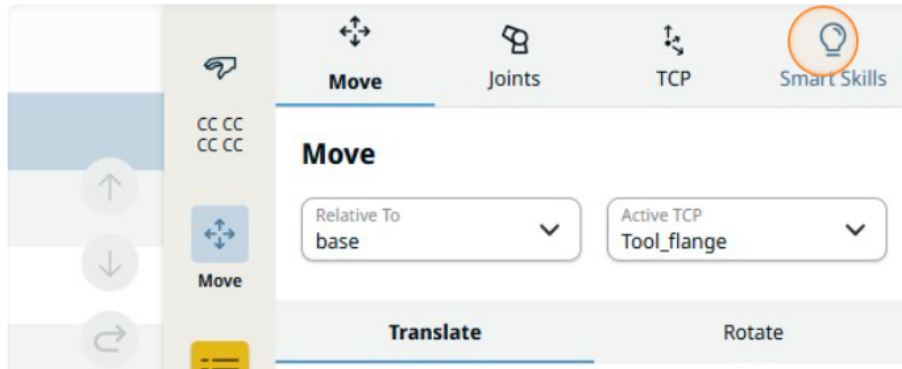


Fig. 6-3: Move Section

Step 4: In the "Smart Skills" menu, select "Gripper Open". The gripper will immediately open.

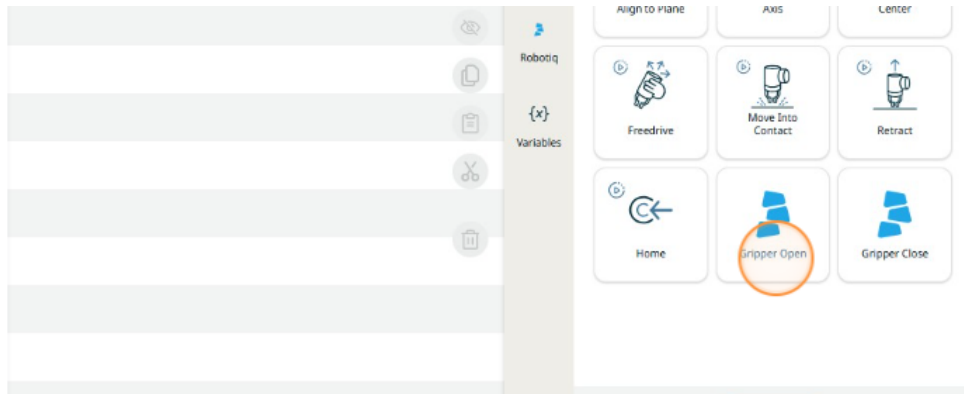


Fig. 6-4: Smart Skills Menu - Gripper Open

Step 5: In the "Smart Skills" menu, select "Gripper Close". The gripper will immediately close.

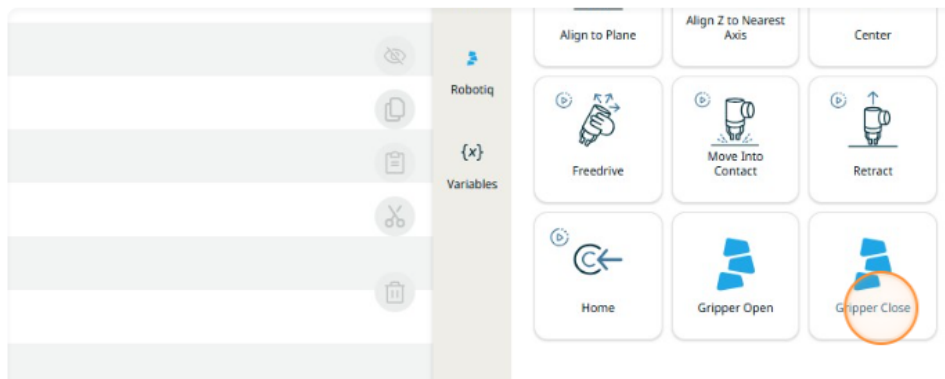


Fig. 6-5: Smart Skills Menu - Gripper Close

Step 6: In the “Smart Skills” menu, select "Teach mode".

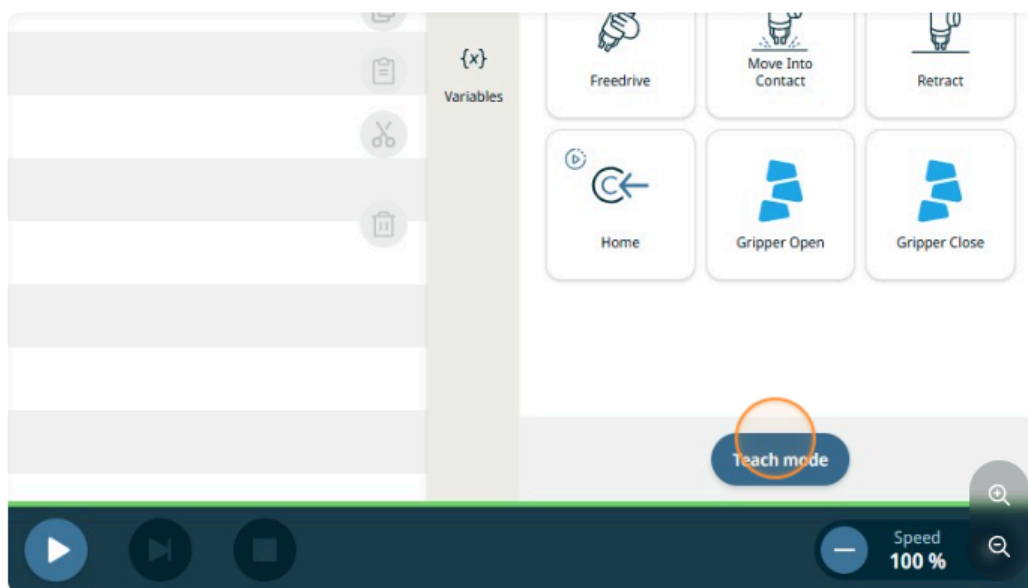


Fig. 6-6: Smart Skills Menu - Teach Mode Button

Note that the Teach Mode status is displayed at the top right of the Smart Skills section of the sidebar.

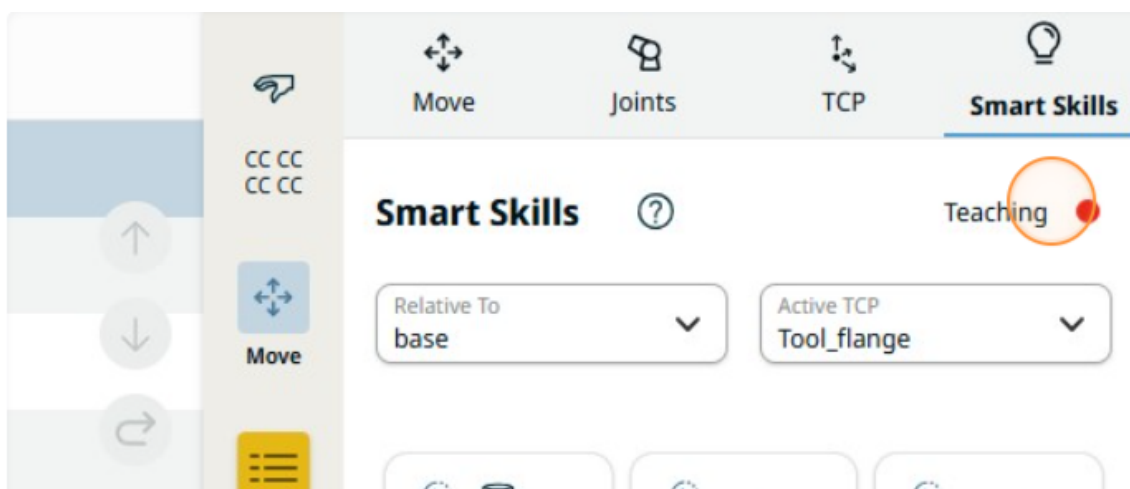


Fig. 6-7: Smart Skills Menu - Teach Mode Status

Step 7: In "Teach Mode", choose "Gripper Open". This action automatically inserts a gripper move instruction to open the gripper into the program tree. You can then customize this gripper move program node as needed.

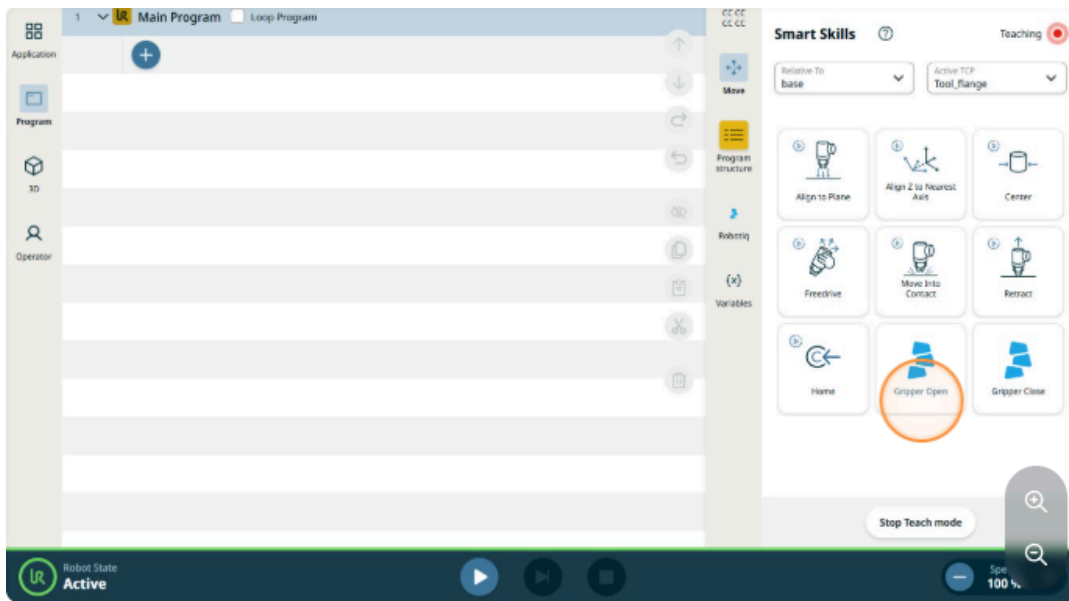


Fig. 6-8: Smarts Skills - Insert Gripper Open Move

Step 8: After clicking "Gripper Close", a Gripper Move action to close the gripper will be added to the program tree. This program node can then be edited as needed.

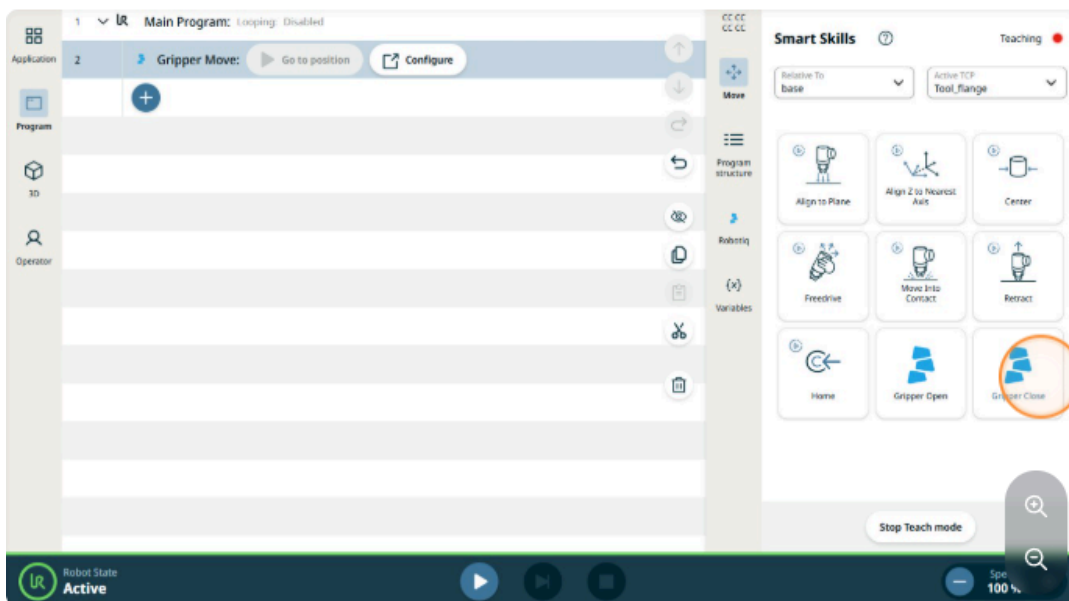


Fig. 6-9 Smarts Skills - Insert Gripper Close Move

b. Modifying the Default Parameters for the Gripper Open and Gripper Close Smart Skills

Step 1: Navigate to the "Application" section.

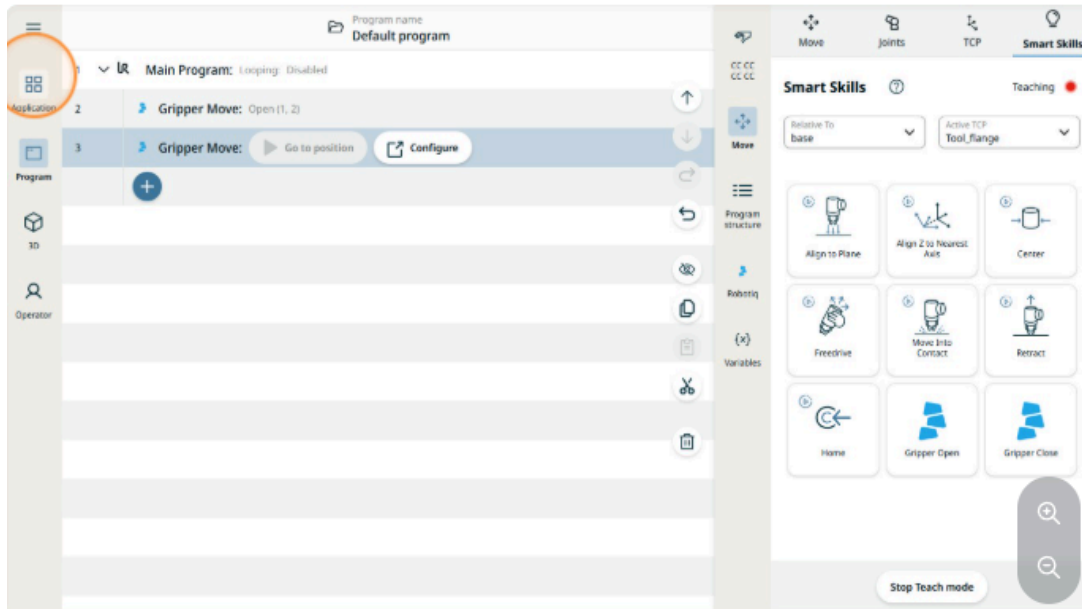


Fig. 6-10: Application Tab - Left Sidebar

Step 2: Under the "Application" menu, select "Smart Skills".

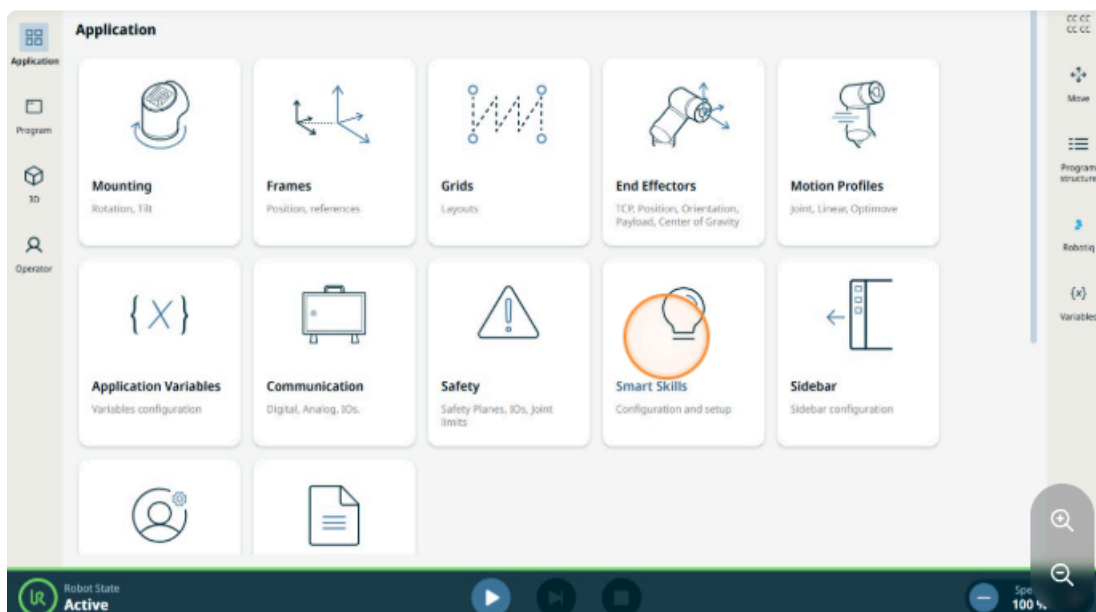


Fig. 6-11: Application Menu - Smart Skills

Step 3: In the “Smart Skills” menu, under Activated, select "Gripper Open" or "Gripper Close".

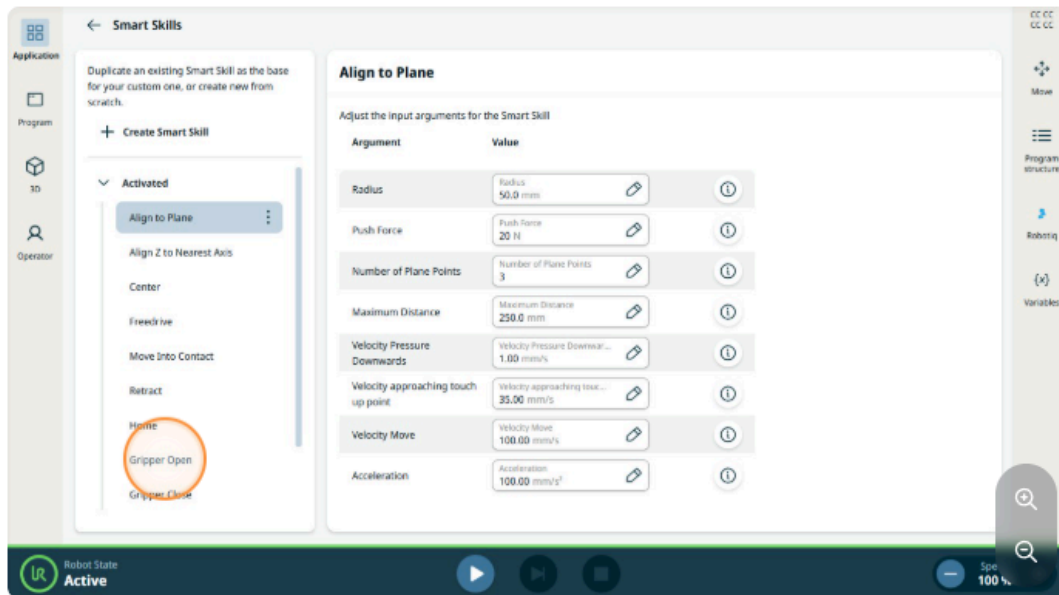


Fig. 6-12: Application - Smart Skills - Gripper Open/Gripper Close

Step 4: By default, both "Gripper Open" and "Gripper Close" Smart Skills will operate all activated grippers. To change this default behavior, uncheck the "Use all activated grippers" box.

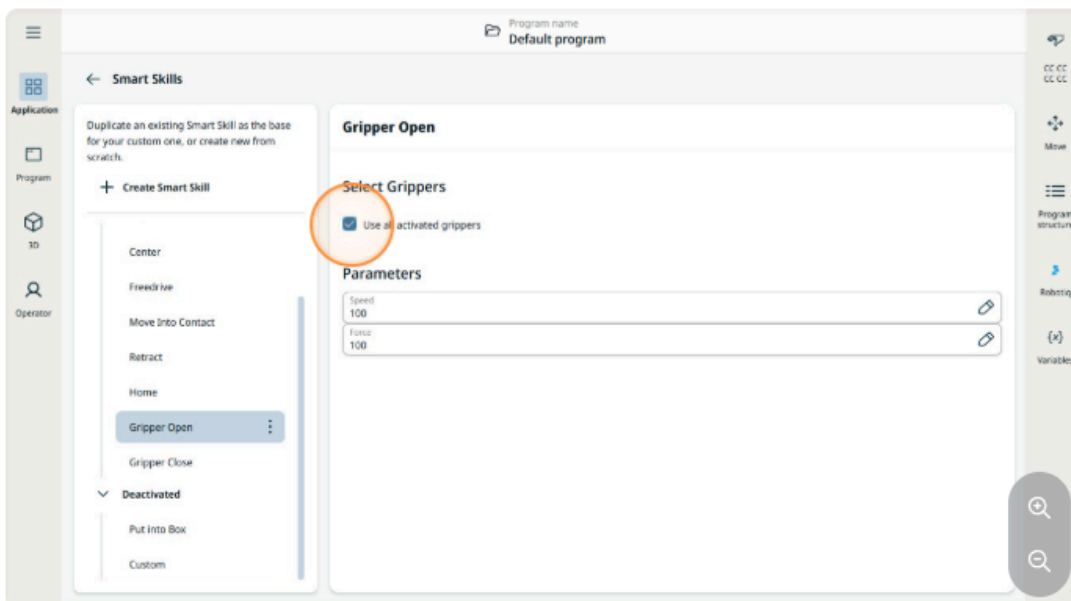


Fig. 6-13: Application - Smart Skills - Gripper Open/Gripper Close - Use all activated grippers

Step 5: If multiple grippers are connected, you will be prompted to manually select the desired gripper. You can also set the default speed and force.

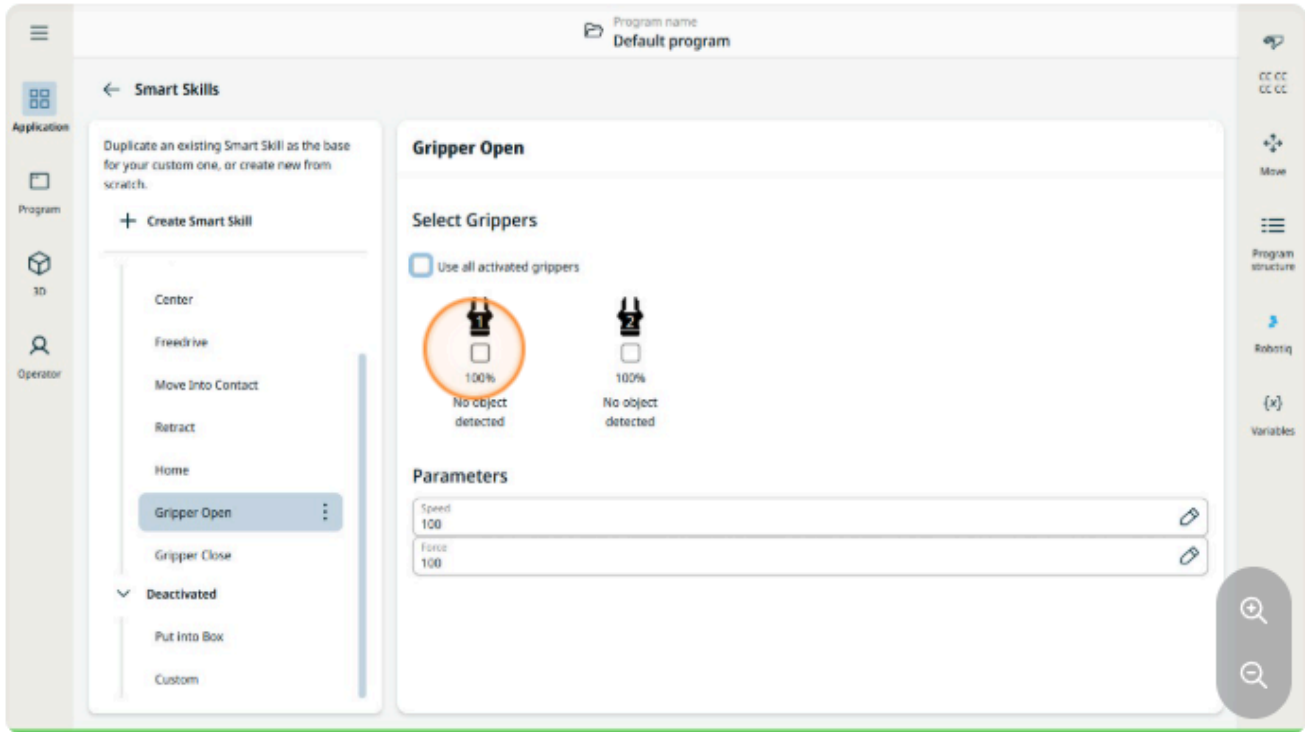


Fig. 6-14: Application - Smart Skills - Gripper Open/Gripper Close - Select Grippers